

Machine Learning - 2301CS621

25-11-2025

Lab - 1

NumPy Exercises

import numpy as np

```
In [2]: import numpy as np
```

```
In [3]: a = np.array([1,2,3,4,5])
```

```
In [4]: print(type(a))
```

```
In [5]: a.dtype
```

```
In [6]: a = np.array(['A', 'B', 'C'])  
a.dtype
```

Create an array of 10 zeros

```
In [7]: np.zeros([10,10])
```

Create an array of 10 ones

```
In [8]: np.ones(10)
```

Create an array of 10 fives

```
In [9]: np.ones(10)*5
```

Create an array of the integers from 10 to 50

```
In [10]: np.arange(10,50,2)
```

Create a 3x3 matrix with values ranging from 0 to 8

```
In [11]: np.arange(0,9,1).reshape(3,3)
```

Create a 3x3 identity matrix

```
In [12]: np.eye(3)
```

Create following matrix =

```
array([[ 1,  2,  3,  4,  5],
       [ 6,  7,  8,  9, 10],
       [11, 12, 13, 14, 15],
       [16, 17, 18, 19, 20],
       [21, 22, 23, 24, 25]])
```

```
In [13]: arr = np.arange(1,26,1).reshape(5,5)
arr
```

Output should be =

```
array([[12, 13, 14, 15],
       [17, 18, 19, 20],
       [22, 23, 24, 25]])
```

```
In [14]: arr[2::,1::]
```

Output should be =

```
array([[ 2],
       [ 7],
       [12]])
```

```
In [15]: arr[0:3:,1:2:]
```

Find Odd Numbers

Extract all odd numbers from `arr` .

```
arr = np.array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

Use Conditional Selection

```
In [16]: arr = np.array([0,1,2,3,4,5,6,7,8,9])
arr[arr%2!=0]
```

Stacking

Stack array `a` and `b` vertically.

```
a = np.arange(10).reshape(2,-1) b = np.repeat(1, 10).reshape(2,-1)
```

```
In [17]: a=np.arange(10).reshape(2,-1)
b=np.repeat(1,10).reshape(2,-1)
np.stack([a,b])
```

Range Filtering

Get all items between 5 and 10 from `a` .

```
a = np.array([2, 6, 1, 9, 10, 3, 27])
```

```
In [18]: a=np.array([2,6,1,9,10,3,27])
a[(a>=5) & (a<=10)]
```

Matrix Multiplication

Multiply a 5x3 matrix by a 3x2 matrix (dot product).

```
In [19]: A = np.random.random((5,3))
B = np.random.random((3,2))
C = np.dot(A,B)
C
```

Context: You are a Data Analyst at "TechCorp". HR has asked you to analyze employee data to understand attrition, salary trends, and job satisfaction.

Instructions:

1. Write your code in the empty cells.

1. Load & Inspect

Load `employee_data.csv` into a DataFrame named `df`. Display the first 5 rows and check column data types.

```
In [38]: import pandas as pd
```

```
In [39]: df = pd.read_csv('employee_data.csv')
df
```

```
In [40]: df.columns
```

2. Filter IT Department

Create a DataFrame `df_rd` containing only employees from the 'Research & Development' department.

```
In [30]: df_rd = df[df['Department']=="Research & Development"]
df[df_rd]
```

3. Salary Check

Find the employee with the highest `MonthlyIncome`. Display their `JobRole` and `MonthlyIncome`.

```
In [31]: temp = df.MonthlyIncome.max()
temp2 = df[df['MonthlyIncome'] == temp]
print(temp2['JobRole'],temp)
```

4. Missing Values

Check if there are any missing values (`NaN`) in the dataset.

```
In [32]: df.isnull().sum()
```

5. Average Age by Role

Calculate the average `Age` for each `JobRole`.

```
In [33]: df=df.groupby(by='JobRole')['Age'].mean()
df
```

6. Attrition Count

How many employees have left the company? (Count values in the `Attrition` column).

```
In [41]: attrition_count = df[df['Attrition'] == 'Yes'].shape[0]
attrition_count
```

7. High Performers

Filter for employees with `PercentSalaryHike` > 15 **AND** `PerformanceRating` == 4.

```
In [42]: df[(df["PercentSalaryHike"]>15) & (df['PerformanceRating']==4)]
```

8. Satisfaction Analysis (Pivot Table)

Create a pivot table showing the **average** `MonthlyIncome` for each `Department` (rows) broken down by `JobSatisfaction` (columns).

```
In [43]: df.pivot_table(index='Department', columns='JobSatisfaction', values='MonthlyIncome')
```

9. Feature Engineering (Seniority)

Create a new column `Seniority`.

- If `YearsAtCompany` > 10, value is `'Senior'`
- Otherwise, value is `'Junior'`

Count how many Seniors vs Juniors there are.

```
In [ ]: df['Seniority']=df.YearsAtCompany.apply(lambda x: 'Senior' if x>10 else 'Junior')
df
```

10. Retention Risk (Complex Filter)

Identify 'High Risk' employees who meet **ALL** these criteria:

1. `MaritalStatus` is 'Single'
2. `JobSatisfaction` is less than 3
3. `DistanceFromHome` > 10

Display their `EmployeeID` and `JobRole`.

```
In [ ]: df[
    (df['MaritalStatus'] == 'Single') &
    (df['JobSatisfaction'] < 3) &
    (df['DistanceFromHome'] > 10)
][['EmployeeID', 'JobRole']]
```